

Settings tested: step from rest to 460 RPM, mean of repeats

ID	What changed	Kp	Ki	Filter α	PWM jitter (% duty)	vs baseline	Overshoot (%)	Settling to ± 5 % (s)	RPM std (RPM)	RPM std vs baseline	runs
A	Baseline (original design)	1.56	41.2	1.0 (off)	0.126	—	1.3	0.14	3.11	—	2
C1	Kp 1.2	1.20	41.2	1.0 (off)	0.096	−24 %	1.2	0.11	2.86	−8 %	2
C2	Kp 1.0	1.00	41.2	1.0 (off)	0.084	−33 %	1.2	0.08	2.94	−6 %	2
C3	Kp 0.8	0.80	41.2	1.0 (off)	0.073	−42 %	1.6	0.08	3.05	−2 %	2
B1	α 0.5	1.56	41.2	0.5	0.075	−41 %	15.3	0.18	3.43	+10 %	2
B2	α 0.3	1.56	41.2	0.3	0.047	−63 %	30.2	0.27	3.27	+5 %	2
B3	α 0.15	1.56	41.2	0.15	0.023	−82 %	65.4	0.62	2.79	−10 %	2
D1	Kp 1.2, α 0.5	1.20	41.2	0.5	0.059	−53 %	16.3	0.14	3.30	+6 %	2
D2	Kp 1.2, α 0.3	1.20	41.2	0.3	0.038	−70 %	30.1	0.28	3.12	≈ 0 %	2
D3	Kp 1.0, α 0.5	1.00	41.2	0.5	0.050	−61 %	12.5	0.13	3.13	+1 %	2
D4	Kp 1.0, α 0.3	1.00	41.2	0.3	0.031	−75 %	34.3	0.32	2.90	−7 %	2

PWM jitter = RMS sample-to-sample change of the PWM command over the last 40 % of the run. Blue row = baseline. Row colour = startup overshoot: green ≤ 5 %, amber 5–15 %, red > 15 %.